بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

LIVE ONLINE COURSE | COHORT FALL 2026

# AI & Machine Learning for Civil Engineers

SESSION 05 · BINARY CLASSIFICATION FOR SLOPE STABILITY

## From Prediction to Engineering Decision: Classification

**Stable or unstable?** When the answer is a category, the model, the metrics and the errors that matter all change.

NUMBER → PROBABILITY → CLASS → DECISION

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WHERE ARE WE?

# From predicting a number to making a decision

SESSIONS 03–04 · REGRESSION

## How strong will this concrete be?

$\hat{y} = 42.7$ MPa

linear regression · trees & forests · MAE · RMSE · $R^2$

The answer is a number on a continuous scale.

TODAY · SESSION 05 · CLASSIFICATION

## Is this slope stable or unstable?

unstable · p = 0.88

logistic regression · trees & forests · confusion matrix

The answer is a category, often with a probability, and the errors are not equally costly.

Same workflow as before. The target is now a class, so the model output and the metrics change.

THE COURSE AT A GLANCE

# Eight sessions. One engineering workflow.

## PART 1 · FOUNDATIONS

**01** From Engineering Problems to Python DONE
AI/ML basics, Google Colab, NumPy & pandas

**02** From Raw Engineering Data to an ML-Ready Dataset DONE
EDA, data quality, outliers, train/test split, leakage

**03** Your First Engineering Prediction Model DONE
Linear regression, error metrics, cross-validation

**04** Nonlinear Engineering Problems: Trees & Ensembles DONE
Decision trees, Random Forest, boosting

## PART 2 · PREDICTION TO DECISION

**05** From Prediction to Engineering Decision: Classification TODAY
Logistic regression, confusion matrix, precision & recall

**06** Artificial Neural Networks
Architecture, activations, training, overfitting

**07** Can We Trust Our ML Model?
Feature importance, SHAP, engineering judgement
Instructor's research area

**08** From Prediction to Probability
Monte Carlo, ML surrogates, probability of failure
Instructor's research area

TODAY · SESSION 05

# Classification, then slope stability

**01**

## The language of classification

Terms · logistic vs linear regression · sigmoid · confusion matrix · recall & precision

**02**

## The slope-stability problem

616 Morgenstern–Price cases · the FOS label · imbalance · three models · stratified CV

**03**

## Results and the engineering decision

Test metrics · confusion matrices · thresholds · which model, and why

### BY THE END OF TODAY YOU CAN

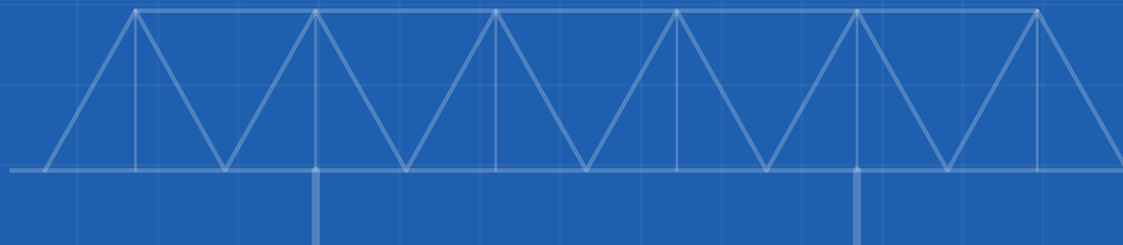
- Explain how logistic regression turns inputs into a probability and a class
- Read a confusion matrix and choose recall, precision or F1 for the task
- Define an engineering label and handle class imbalance correctly
- Select a model from the cost of its errors, not from accuracy alone

# 01

- a Regression or classification?
- b The vocabulary of classification
- c From linear to logistic regression
- d Score → probability → class
- e The confusion matrix and its metrics

PART 01 · CLASSIFICATION FUNDAMENTALS

# The language of classification



THE TARGET DECIDES, EVERY TIME

# Same inputs, different question

1.37 FOS

REGRESSION · SESSIONS 03–04

## Predict a number

From infinitely many possible values.

| | |
|---|---|
| Factor of safety of a slope | 1.37 |
| Settlement of a footing | 24 mm |
| Concrete strength | 42.7 MPa |

Stable Unstable

CLASSIFICATION · TODAY

## Predict a category

From a small, fixed set of outcomes.

| | |
|---|---|
| Slope: FOS below 1.0? | unstable |
| Settlement within the limit? | acceptable |
| Steel plate fault type | 1 of 7 |

**A continuous result can become a class through an engineering threshold, such as FOS < 1.0.**

WORDS WE WILL USE ALL SESSION

# The vocabulary of classification

## Class · label

One of the possible categories. The training table stores it as y.

`stable = 0 · unstable = 1`

## Positive class

The class we want to detect. It defines what recall and precision mean.

`positive = unstable`

## Binary · multiclass

Two classes, or three and more mutually exclusive classes.

`2 slope classes · 7 steel faults`

## Probability

The model's estimate that a case belongs to the positive class.

`predict_proba() → 0.88`

## Decision threshold

The probability above which we call a case positive.

`p ≥ 0.50 → unstable`

## Class imbalance

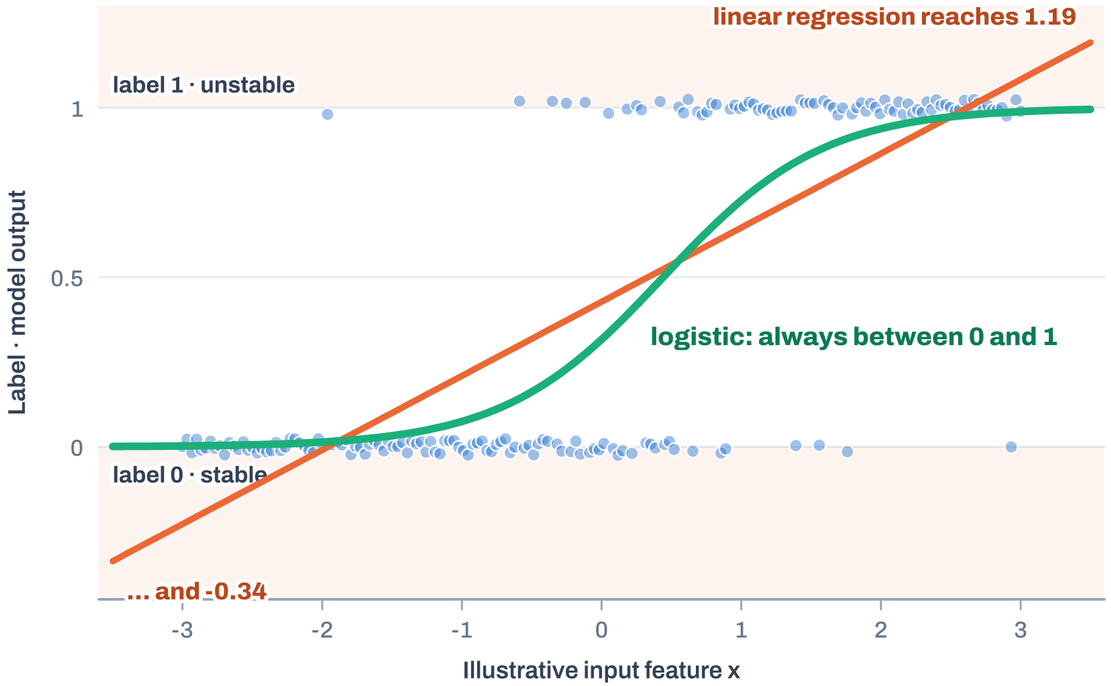
One class is much rarer than the other. It changes which metrics matter.

`93 unstable of 616`

LINEAR REGRESSION VS LOGISTIC REGRESSION · 1 OF 2

# Why not fit linear regression to 0/1 labels?

## Same binary labels, two models




### LINEAR REGRESSION

Predicts values from **−0.34** to **1.19**: not valid probabilities, and a straight line ignores the S-shape.

### LOGISTIC REGRESSION

Squeezes the same linear score through the sigmoid: every output lies between 0 and 1.

### WHAT WE WANT

**P(unstable | β, c/(γH), φ)**

Illustrative one-feature data from the session notebook (180 points, random seed 42); the slope dataset has three inputs.

LOGISTIC REGRESSION IN FOUR STEPS

# Inputs → score → probability → class

**1**

## Inputs

$\beta$, $c/(\gamma H)$, $\varphi$

Slope angle, cohesion term, friction angle.

**2**

## Linear score

$z = b_0 + \Sigma\, b_j x_j$

Any real number. Same form as linear regression.

**3**

## Probability

$p = 1 / (1 + e^{-z})$

The sigmoid maps z into (0, 1).

**4**

## Decision

$p \geq$ threshold ?

Yes → unstable · No → stable.

**“Regression” stays in the name because the model is linear on the log-odds scale. The class comes from the threshold.**

STEP 1 · BUILD A LINEAR SCORE

# A weighted sum of the inputs, as in Session 03

FOR THE SLOPE-STABILITY DATASET

$$z = b_0 + b_1\beta + b_2[c/(\gamma H)] + b_3\varphi$$

## $\beta$

### Slope angle

Steepness of the slope face.

15° – 65° · 11 levels

## $c/(\gamma H)$

### Cohesion term

Dimensionless stability number: cohesion over unit weight × height.

0 – 0.30 · 7 levels

## $\varphi$

### Friction angle

Shear-strength parameter of the soil.

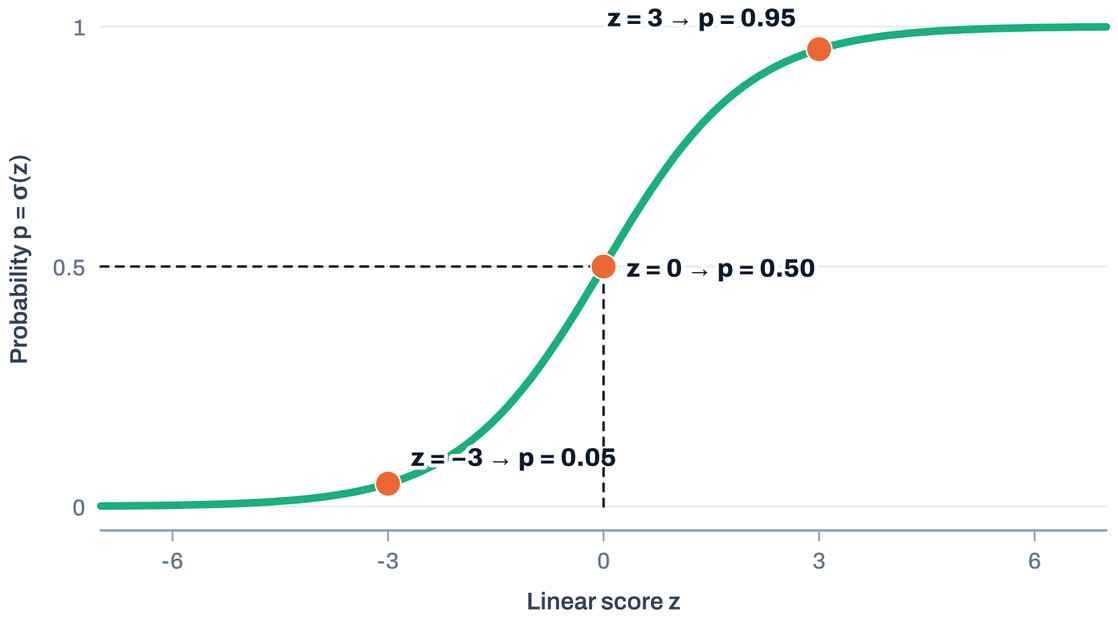
10° – 45° · 8 levels

z can be any number. The coefficients b are learned from the training cases; their sizes depend on units and scaling.

STEP 2 · MAP THE SCORE TO A PROBABILITY

# The sigmoid squashes any score into (0, 1)

**p = σ(z)**




THE SIGMOID

$$p = \frac{1}{1 + e^{-z}}$$

---

`z < 0` → p < 0.5

`z = 0` → p = 0.5

`z > 0` → p > 0.5

A model probability is an estimate. It is not automatically calibrated, and it is not a physical probability of failure.

STEP 3 · WHAT THE COEFFICIENTS MODEL

# Linear in the log-odds

**PROBABILITY**

$$p \in (0, 1)$$

`p = 0.80`

Bounded, so a straight line cannot model it directly.

**ODDS**

$$\frac{p}{1-p} \in (0, \infty)$$

`0.80 / 0.20 = 4 : 1`

“Four times as likely to be unstable as stable.”

**LOG-ODDS · LOGIT**

$$\log \frac{p}{1-p} = z$$

`log 4 = 1.39`

Unbounded in both directions, so a linear model can describe it.

**A coefficient is the change in log-odds per unit of an input, others held fixed. It is an association in this numerical dataset, not a physical cause.**

STEP 4 · PROBABILITY BECOMES A CLASS

# A threshold turns p into a decision

default rule: unstable if p ≥ 0.50




**LOWER THE THRESHOLD**

More cases flagged unstable: fewer missed unstable slopes, more false alarms.

**RAISE THE THRESHOLD**

Fewer cases flagged: fewer false alarms, more missed unstable slopes.

**0.50 is a software default, not a law of soil mechanics. Choose it on validation data, never on the test set.**

A WORKED PREDICTION · THE FITTED MODEL, A REAL TEST CASE

# From inputs to decision, with real numbers

**1 · INPUTS**

| | |
|---|---|
| β | 45° |
| c/(γH) | 0.05 |
| φ | 10° |

**2 · LINEAR SCORE**

$z = 1.15 + 0.064 \cdot \beta - 24.5 \cdot c/(\gamma H) - 0.087 \cdot \varphi$

$z = 1.95$

**3 · PROBABILITY**

$p = 0.88$

$1 / (1 + e^{-1.95})$

**4 · DECISION**

**unstable**

$p \geq 0.50$

CHECK Morgenstern–Price gives FOS = 0.66 for this slope: truly unstable, and correctly flagged.

Coefficients of the tuned logistic regression (C = 0.1, class_weight = "balanced"), converted from the scaled model to original units. Class weighting shifts the probabilities: read them as scores, not calibrated chances of failure.

LINEAR REGRESSION VS LOGISTIC REGRESSION · 2 OF 2

# The essential distinction

| QUESTION | Linear regression · Session 03 | Logistic regression · today |
|---|---|---|
| **Typical target** | Continuous quantity | **Class label, often 0 / 1** |
| **The linear part** | Predicts the response directly | **A score on the log-odds scale** |
| **Output** | Any real number | **Probability in (0, 1), via the sigmoid** |
| **Fitting objective** | Minimise squared errors | **Maximise likelihood · minimise log loss** |
| **Threshold** | Usually not needed | **Turns a probability into a class** |
| **Evaluation** | MAE · RMSE · $R^2$ | **Confusion matrix · recall · precision · F1** |

**With a fixed threshold, logistic regression draws a straight boundary in the input space, unless we transform the features.**

POSITIVE CLASS = UNSTABLE (FOS < 1.0)

# The confusion matrix: where every prediction lands

| | PREDICTED: STABLE | PREDICTED: UNSTABLE |
|---|---|---|
| ACTUAL: STABLE | **TN**<br>**True negative**<br>Stable, predicted stable. Correct. | **FP**<br>**False positive · false alarm**<br>Stable, flagged unstable. Costs an unnecessary check. |
| ACTUAL: UNSTABLE | **FN**<br>**False negative · missed warning**<br>Unstable, predicted stable. Potentially consequential. | **TP**<br>**True positive**<br>Unstable, flagged unstable. Detected. |

**HOW TO READ IT**

Rows are the truth, columns are the prediction. The diagonal is correct.

**FOR SAFETY SCREENING**

Inspect the **bottom-left cell** first: unstable slopes the model called stable.

DEFINITIONS BEFORE SCORES

# Four metrics, four different questions

RECALL (UNSTABLE) · SENSITIVITY

$$\text{Recall} = \frac{TP}{TP + FN}$$

Of all unstable cases, what fraction did we detect?

PRECISION (UNSTABLE)

$$\text{Precision} = \frac{TP}{TP + FP}$$

Of all cases flagged unstable, what fraction really were?

F1 SCORE

$$F1 = \frac{2 \cdot P \cdot R}{P + R}$$

One number that balances precision and recall (harmonic mean).

BALANCED ACCURACY

$$BA = \frac{\text{Recall}_{\text{unstable}} + \text{Recall}_{\text{stable}}}{2}$$

Gives both classes equal weight, unlike ordinary accuracy.

**Our priority today: do not miss unstable slopes. So recall for the unstable class comes first, with precision in view.**

WHY ACCURACY ALONE CAN MISLEAD

# A model that never warns can still look “accurate”

ALWAYS PREDICT “STABLE” · 124 TEST SLOPES

| | PRED. STABLE | PRED. UNSTABLE |
|---|---|---|
| STABLE | 105 TN | 0 FP |
| UNSTABLE | 19 FN · all missed | 0 TP |

ACCURACY

**84.7%**

It looks good, because 85% of slopes are stable anyway.

RECALL FOR UNSTABLE

**0 of 19**

It detects none of the unstable slopes. Balanced accuracy is 0.50: no better than chance.

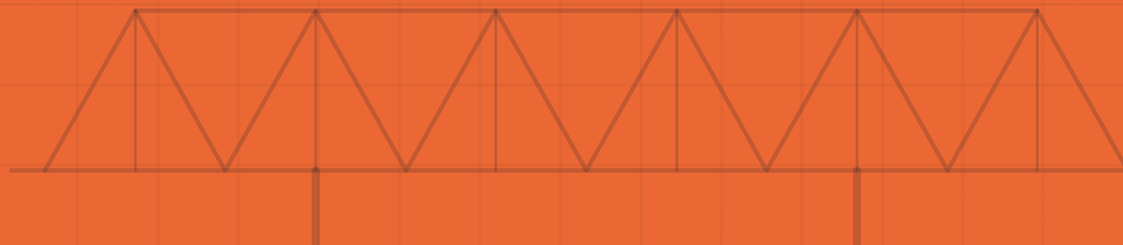
**With imbalanced classes, read the confusion matrix and the recall of the rare class, not accuracy alone.**

# 02

- a The engineering question
- b 616 Morgenstern–Price cases
- c The FOS label and class imbalance
- d The stability boundary
- e Three models, stratified CV

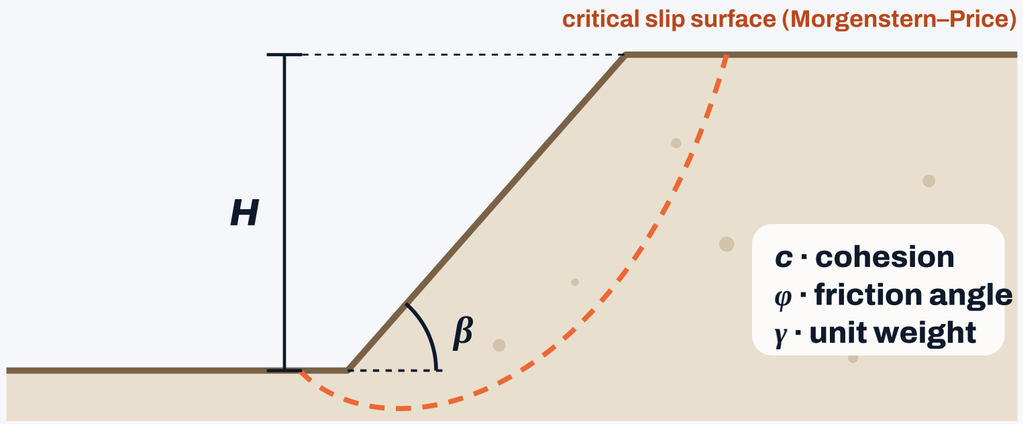
PART 02 · THE SLOPE-STABILITY PROBLEM

# From factor of safety to a classification



THE ENGINEERING QUESTION

# Can three inputs classify a slope as stable or unstable?




## Can β, c/(γH) and φ classify cases below the limiting-equilibrium threshold?

**INPUTS X**

Slope angle β (°), dimensionless c/(γH), friction angle φ (°).

**THE LABEL Y**

From the factor of safety of a Morgenstern–Price analysis:

unstable = 1 · FOS < 1.0

stable = 0 · FOS ≥ 1.0

The labels reproduce a numerical analysis. They are not independent field failures.

THE INSTRUCTOR'S DATASET · FIRST AUDIT, AS IN SESSION 02

# 616 cases on a structured grid

**616**
slope cases, one Morgenstern–Price analysis each

**11×7×8**
levels of β × c/(γH) × φ: every combination once

**0**
missing values · 0 repeated input combinations

**0.17–7.43**
range of the factor of safety

## DATA-QUALITY DECISIONS

**itr** is a row identifier: never a predictor

Blank helper columns in the spreadsheet are dropped

Slip-surface columns (Xmp, Ymp, Rmp) are outputs of the analysis: using them would leak the answer

A grid means test cases sit between familiar training levels

Session_05_Integrated…Slope_Stability.ipynb audit

```
df = raw[["itr", "Beta", "CbygammaH", "phi", "fmp"]]
df.shape, df.isna().sum().sum()
df.duplicated(["beta_deg", "c_by_gammaH", "phi_deg"]).sum()
```

OUTPUT

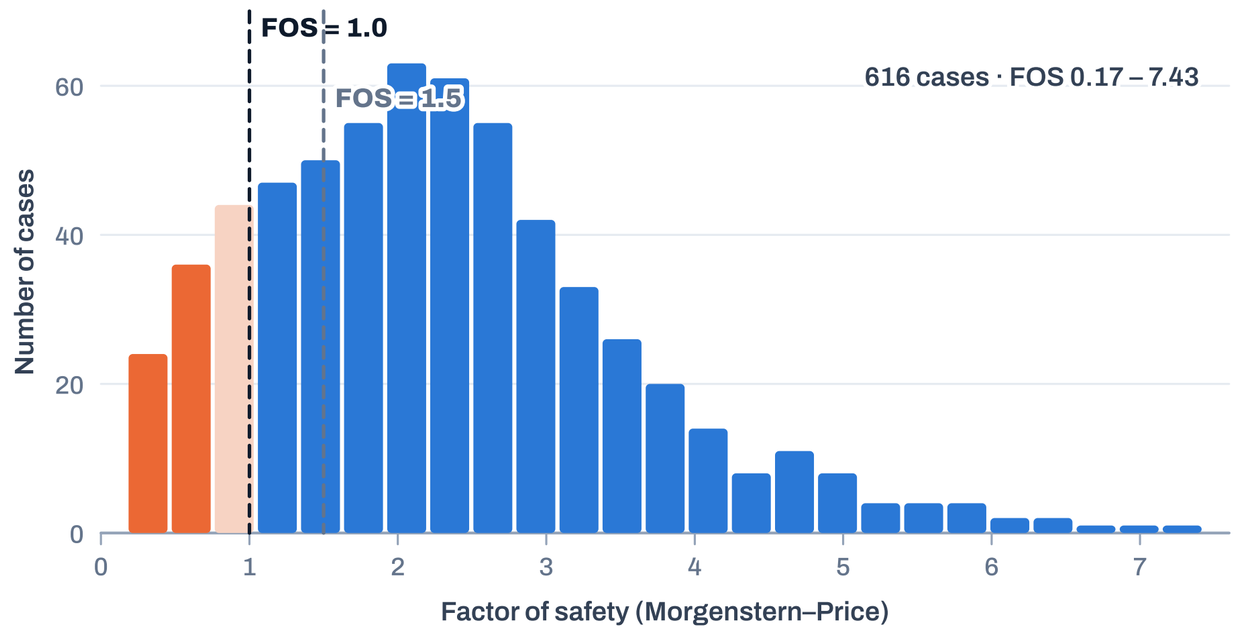
```
((616, 5), 0)
0
```

THE LABEL IS AN ENGINEERING CHOICE

# Where we cut the factor of safety defines the problem

## Factor of safety of all 616 cases

Orange: below 1.0. No case has FOS exactly 1.0.




| LABEL THRESHOLD | BELOW | AT / ABOVE |
|---|---|---|
| FOS < 1.0 | 93 | 523 |
| FOS < 1.5 | 181 | 435 |

**1.0 · OUR PRIMARY LABEL**

The limiting-equilibrium threshold.

**1.5 · A DIFFERENT QUESTION**

A design or screening threshold, not automatically physical failure.

PRIMARY LABEL: UNSTABLE = FOS < 1.0

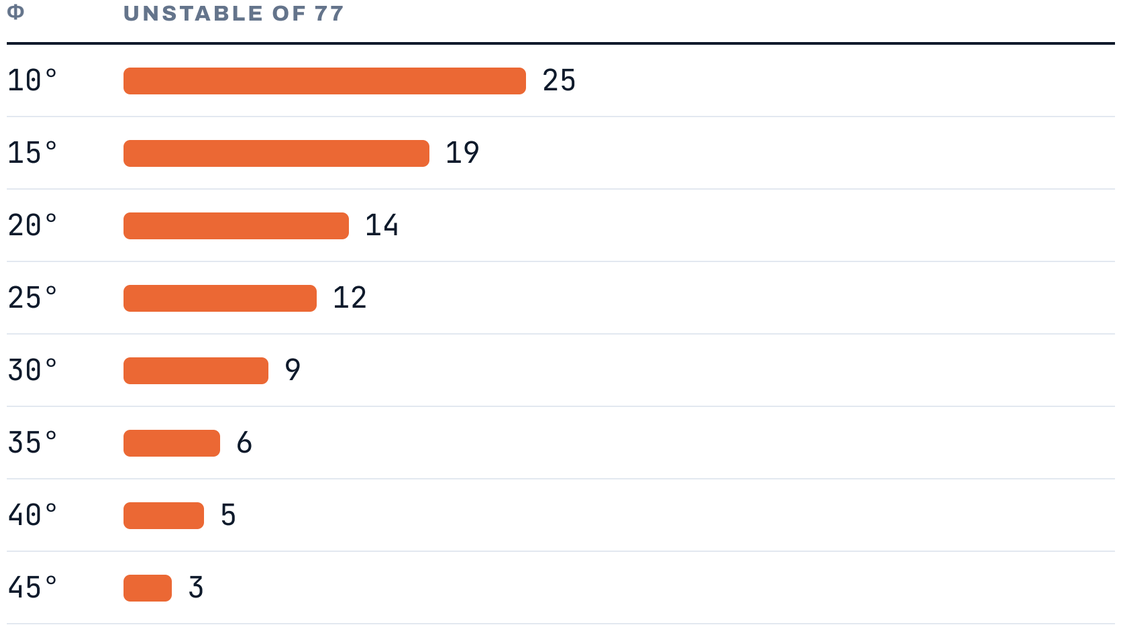
# Class imbalance: unstable slopes are the minority

**Stable** 523 · 84.9%

**Unstable** 93 · 15.1%

Enough for a teaching exercise, but the minority-class metrics need careful reading. Imbalance is not a data defect: it changes which metrics matter.

UNSTABLE CASES AT EACH FRICTION ANGLE



TWO DIFFERENT THRESHOLDS: DO NOT CONFUSE THEM

# One defines the target. One defines the decision.

THE FOS LABEL THRESHOLD

## What “unstable” means

**FOS < 1.0 → y = 1**

Set before modelling, from engineering meaning

Changing it changes the target and the class balance

THE PROBABILITY THRESHOLD

## When the model says “unstable”

**$p \geq 0.50$ → predict 1**

Set after modelling, on validation data

Changing it trades precision against recall for the same target

EXAMPLE

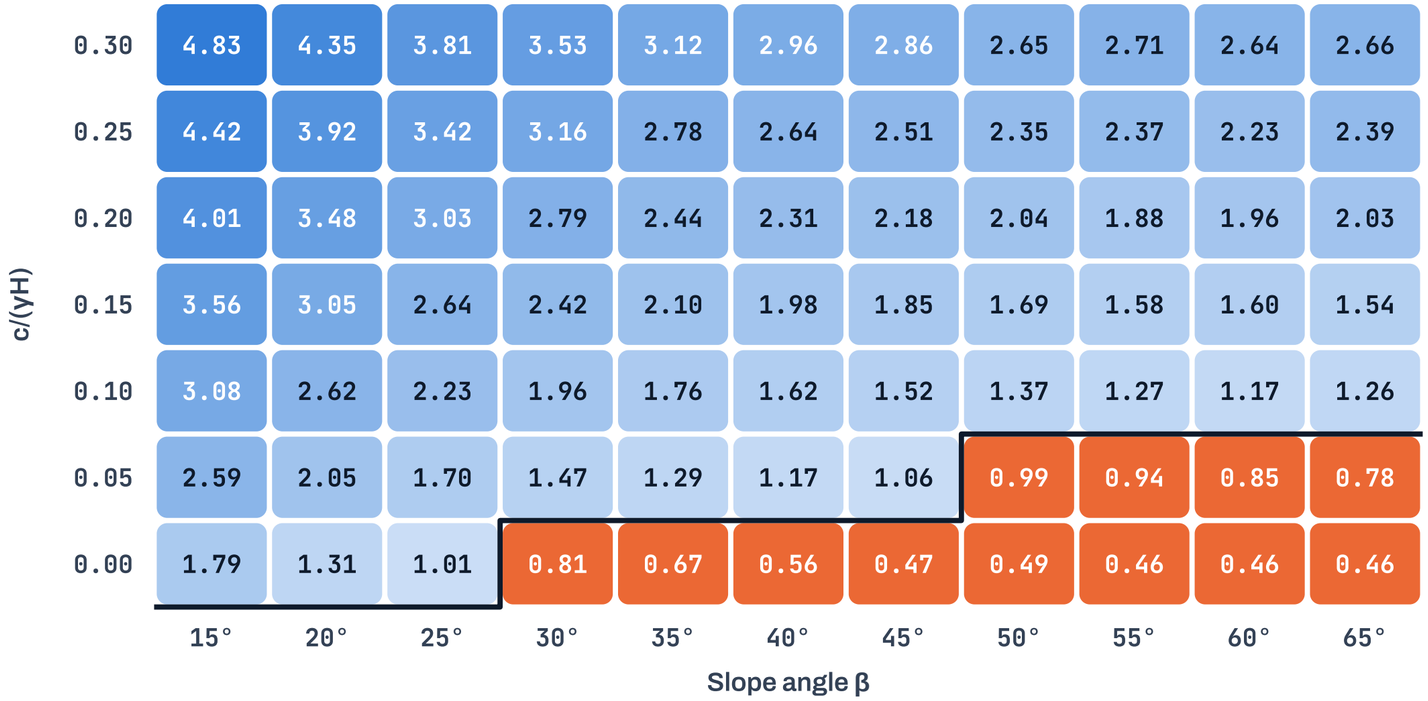
A slope with **FOS = 0.95** is labelled unstable (y = 1). The model gives **p = 0.42**. At threshold 0.50 it predicts stable: **a false negative**. At 0.30 it would be flagged.

THE STABILITY BOUNDARY IS NOT A SINGLE CUT-OFF

# FOS on one slice of the grid: φ = 25°

## Factor of safety for every β and c/(γH) at φ = 25°

Orange: FOS < 1.0 (unstable). The black line is the FOS = 1 boundary in this slice.




**READ IT**

The boundary depends on β and c/(γH) **together**: steeper slopes need more cohesion.

**THIRD INPUT**

φ shifts the whole surface: from 25 unstable cases at φ = 10° to 3 at φ = 45°.

A slice is illustrative: the real boundary lives in three dimensions.

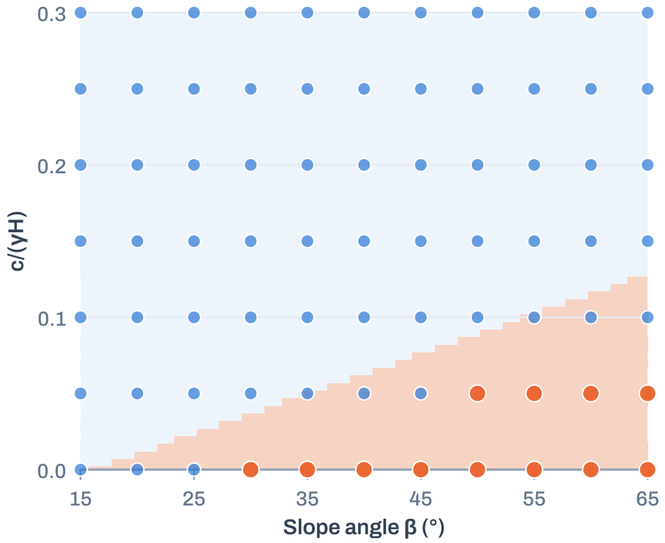
THREE MODELS, THREE INDUCTIVE BIASES · TUNED MODELS ON THE Φ = 25° SLICE

# Each model draws a different kind of boundary

unstable case · stable case · region predicted unstable

## Logistic regression

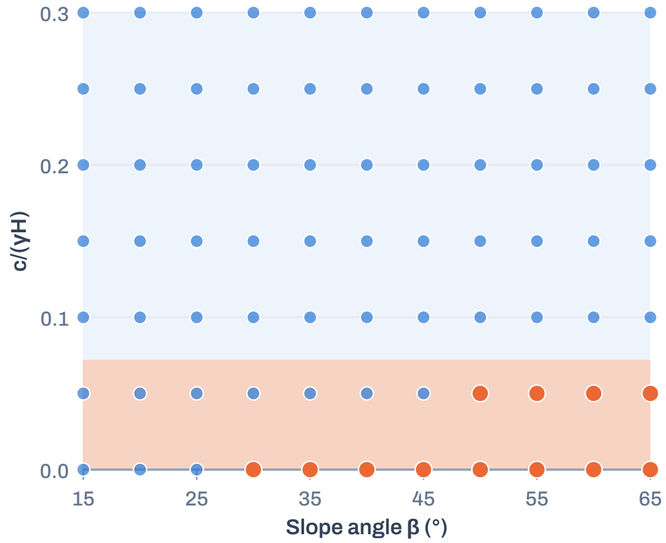
Probability-based; scaling in a pipeline.




A straight boundary

## Decision tree

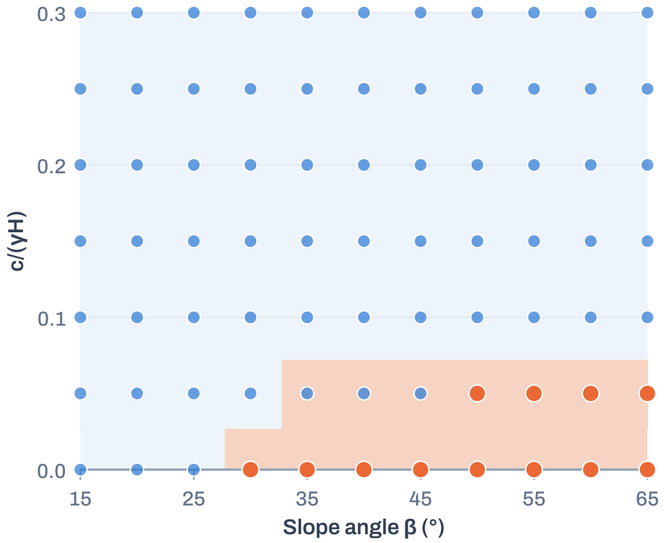
Threshold rules; can overfit.




Axis-aligned steps

## Random forest

Many trees combined; lower variance.




A flexible, stepped boundary

The orange regions reaching stable cases are the models' false alarms. Class weighting makes all three cautious.

WAYS TO HANDLE CLASS IMBALANCE

# Make the rare class count, inside the training data

### Stratify
Keep 15% unstable in the train/test split and in every CV fold.

### Prioritise unstable recall
While watching precision and F1.

### class_weight="balanced"
Errors on the rare class cost more during fitting.

### Adjust the probability threshold
On validation data, never on the test set.

### Resampling, if needed
Only inside each CV training fold.

REAL EFFECT OF CLASS WEIGHTING · LOGISTIC REGRESSION · 5-FOLD CV, TRAINING SET

**No weighting · recall 0.61**

| | PRED. S | PRED. U |
|---|---|---|
| S | 418 | 0 |
| U | 29 missed | 45 |

**Balanced · recall 1.00**

| | PRED. S | PRED. U |
|---|---|---|
| S | 379 | 39 false alarms |
| U | 0 | 74 |

Weighting finds all 74 unstable training cases instead of 45, at the price of 39 false alarms.

SAME DISCIPLINE AS SESSION 03, NOW STRATIFIED

# Stratified split, then CV on training data only

**616 cases**
93 unstable

→

**Training 80% · 492**
418 stable · 74 unstable

**Test 20% · 124**
105 · 19

## RULES

- **stratify=y** keeps 15% unstable in both parts
- 5-fold **StratifiedKFold** on the training set
- **GridSearchCV** tunes each model on the folds
- The test set stays locked until the final evaluation
- On a grid, a random split mostly tests interpolation

Session_05_Integrated…Slope_Stability.ipynb — split · cv

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y)
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
gs = GridSearchCV(model, grid, scoring="recall", cv=cv)
gs.fit(X_train, y_train)
# training data only
```

TUNING STRATEGY · SCORING = RECALL FOR THE UNSTABLE CLASS

# What each model is allowed to try

| MODEL | HYPERPARAMETERS SEARCHED | WHY |
|---|---|---|
| **Logistic regression** | `C ∈ {0.1, 1, 10}`<br>`class_weight ∈ {None, balanced}` | C controls regularisation; scaling sits inside the pipeline |
| **Decision tree** | `max_depth ∈ {2, 3, 4, 5, None}`<br>`min_samples_leaf ∈ {1, 3, 5, 10}` · `class_weight` | Limits on depth and leaf size control overfitting |
| **Random forest** | `max_depth ∈ {None, 4, 8}` · `min_samples_leaf ∈ {1, 3, 5}`<br>`max_features ∈ {sqrt, 1.0}` · `class_weight` · 100 trees | Tree diversity and complexity |

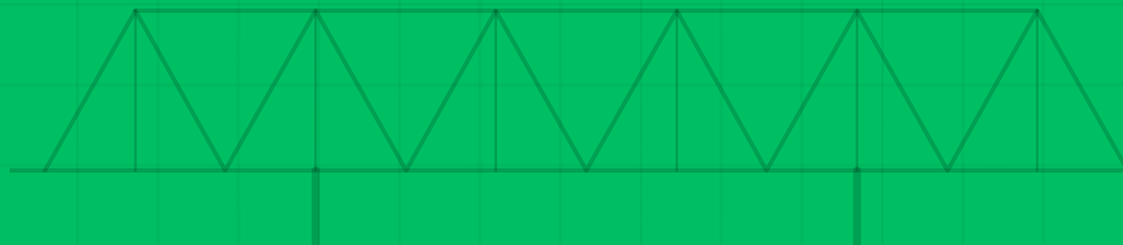
⚠ **Recall-focused tuning can raise false alarms. Always look at precision and the confusion matrix too.**

# 03

- a Baseline and CV results
- b Held-out test performance
- c Confusion matrices
- d Thresholds: the precision–recall trade-off
- e Which model, and what the results do not show

PART 03 · RESULTS & DECISION

# From scores to an engineering decision



FIVE-FOLD CV ON THE TRAINING SET · USED FOR MODEL SELECTION, NOT THE FINAL SCORE

# Baseline first, then the tuned models

## Majority baseline

**0.000**

recall · always “stable”

```
DummyClassifier(
 strategy="most_frequent")
```

## Logistic regression

**1.000**

best mean CV recall

```
C = 0.1
class_weight = "balanced"
```

## Decision tree

**0.987**

best mean CV recall

```
max_depth = 3
min_samples_leaf = 1
class_weight = "balanced"
```

## Random forest

**0.959**

best mean CV recall

```
max_depth = None
min_samples_leaf = 5
max_features = 1.0
class_weight = "balanced"
```

All three tuned models chose class_weight = "balanced". CV picks the settings; only the test set gives the final score.

SELECTED MODELS EVALUATED ONCE ON THE 124 HELD-OUT TEST SLOPES

# Held-out test performance

| MODEL | ACCURACY | BALANCED ACCURACY | PRECISION (UNSTABLE) | RECALL (UNSTABLE) | F1 (UNSTABLE) | ROC-AUC | AVERAGE PRECISION |
|---|---|---|---|---|---|---|---|
| Majority baseline | 0.847 | 0.500 | 0.000 | 0.000 | 0.000 | 0.500 | 0.153 |
| Logistic regression | 0.895 | 0.938 | 0.594 | 1.000 | 0.745 | 0.995 | 0.975 |
| Decision tree | 0.871 | 0.924 | 0.543 | 1.000 | 0.704 | 0.940 | 0.680 |
| Random forest | 0.919 | 0.909 | 0.680 | 0.895 | 0.773 | 0.980 | 0.923 |

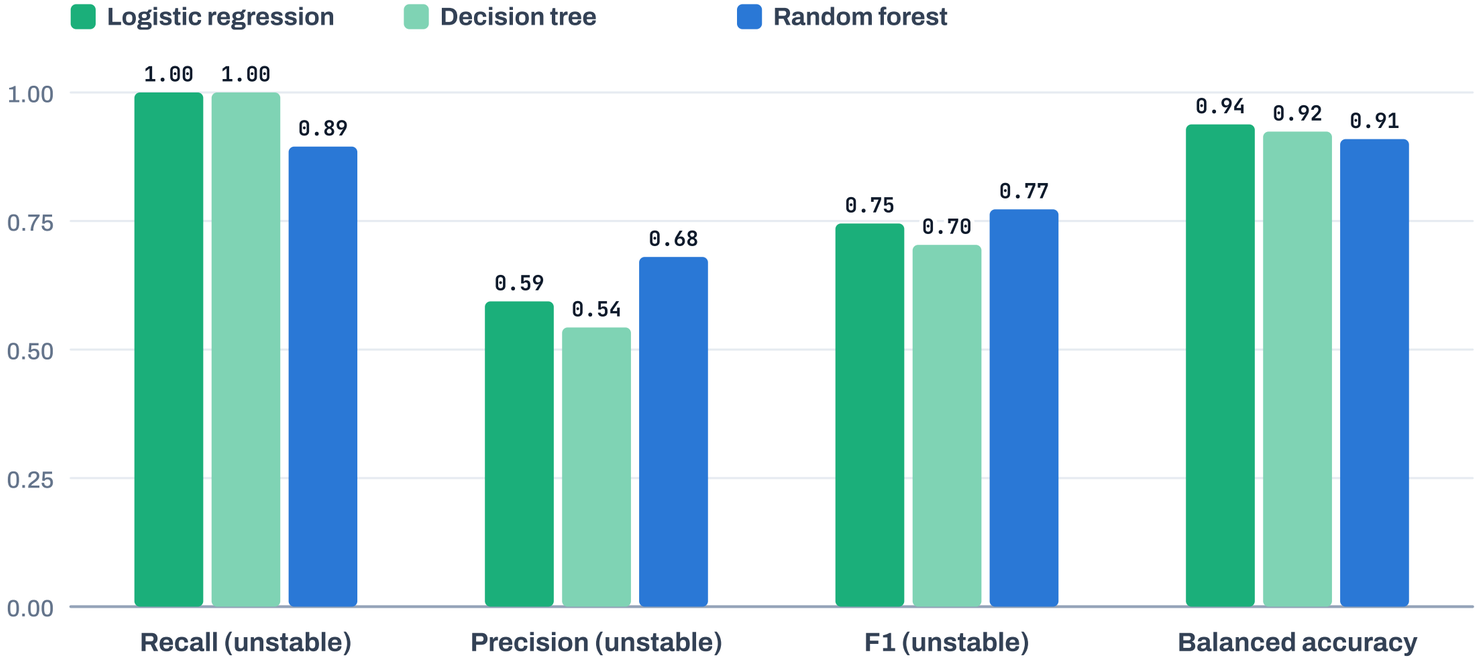
ROC-AUC and average precision judge how well a model ranks cases across all thresholds; the others use the 0.50 threshold.

Test set: 105 stable and 19 unstable slopes. With only 19 unstable cases, one slope changes recall by about 0.05.

COMPARE SAFETY-RELEVANT TEST METRICS

# No model wins on every metric

## Test metrics of the three tuned models




Logistic regression and the tree catch every unstable slope; the forest raises fewer false alarms but misses 2.

ROWS = ACTUAL · COLUMNS = PREDICTED · TEST SET

# Confusion matrices reveal the error types

## Logistic regression

| | PRED. STABLE | PRED. UNSTABLE |
|---|---|---|
| STABLE | 92 TN | 13 FP · false alarm |
| UNSTABLE | 0 FN · missed | 19 TP |

recall 1.000 · precision 0.594

## Decision tree

| | PRED. STABLE | PRED. UNSTABLE |
|---|---|---|
| STABLE | 89 TN | 16 FP · false alarm |
| UNSTABLE | 0 FN · missed | 19 TP |

recall 1.000 · precision 0.543

## Random forest

| | PRED. STABLE | PRED. UNSTABLE |
|---|---|---|
| STABLE | 97 TN | 8 FP · false alarm |
| UNSTABLE | 2 FN · missed | 17 TP |

recall 0.895 · precision 0.680

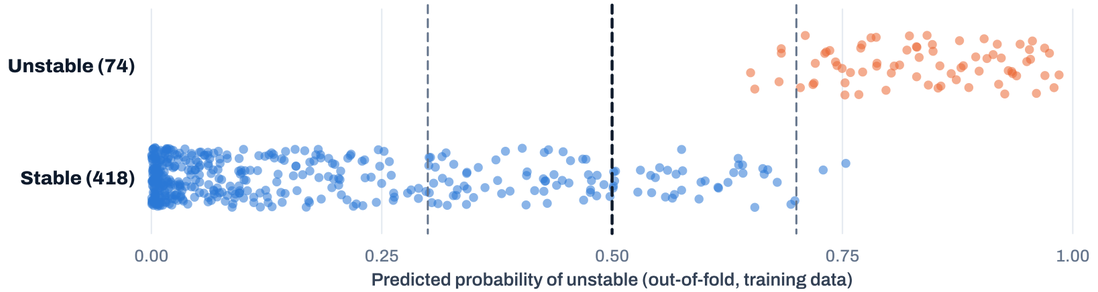
⚠ Focus on the orange cell: actual unstable slopes predicted stable. Random forest: 2. The others: 0.

THRESHOLDS AND PROBABILITIES · LOGISTIC REGRESSION · OUT-OF-FOLD, TRAINING DATA

# Moving the threshold trades recall for precision

## Predicted probability of “unstable” for every training slope

Each case scored by a model that did not see it (5-fold CV). Dashed lines: thresholds 0.3 · 0.5 · 0.7.




| THRESHOLD | 0.3 | 0.5 | 0.7 | 0.9 |
|---|---|---|---|---|
| Recall (unstable) | 1.00 | 1.00 | 0.93 | 0.30 |
| Precision (unstable) | 0.43 | 0.65 | 0.97 | 1.00 |
| Missed · false alarms | 0 · 98 | 0 · 39 | 5 · 2 | 52 · 0 |

Choose the threshold here, on validation data and from the cost of each error. Then evaluate once on the test set.

THE ANSWER DEPENDS ON THE ERROR WE MOST WANT TO AVOID

# Which model do we choose?

PROVISIONAL SCREENING CANDIDATE

## Logistic regression

| 1.000 | 0.594 | 0.745 | 0.938 |
|---|---|---|---|
| recall | precision | F1 | bal. acc. |

Detected every unstable test slope, and is comparatively interpretable. Cost: 13 false alarms.

THE ALTERNATIVE

## Random forest

| 0.895 | 0.680 | 0.773 | 0.909 |
|---|---|---|---|
| recall | precision | F1 | bal. acc. |

Better precision and F1, but it missed 2 unstable slopes. Reasonable if false alarms are costly.

**OUR STATED PRIORITY** Avoid missing unstable slopes → logistic regression, for this split. Not a universal winner: only 19 unstable test cases on a structured grid.

WHAT THESE RESULTS DO, AND DO NOT, SHOW

# A classifier of a numerical method, not of real slopes

### THEY SHOW

- How well classifiers reproduce labels from Morgenstern–Price analyses
- How imbalance, class weights and thresholds change the errors
- A disciplined workflow: baseline, stratified CV, one final test

### THEY DO NOT SHOW

- Independent validation against field failures
- Performance in unseen parameter regions: a random split of a grid tests interpolation
- Real-world slope safety. Decisions still need proper analysis, standards and judgement

**A stronger follow-up test: hold out whole regions of the grid (for example all φ = 45° cases) and predict them.**

HANDS-ON · IN THE NOTEBOOK

# Your turn: six questions to answer

01

Change the FOS label threshold to 1.5 and recompute the class balance. What question does it now answer?

`df.fos < 1.5`

02

Compare the majority baseline with the tuned models. Why is accuracy insufficient?

`DummyClassifier`

03

Which model detects the most unstable test cases? What is its precision?

`classification_report`

04

Explain the false-negative cell of each confusion matrix in engineering terms.

`ConfusionMatrixDisplay`

05

Turn class weighting off. What happens to recall and to false alarms?

`class_weight=None`

06

Design a grid-region holdout. Why is it harder than a random split?

`X[df.phi_deg == 45]`

KEY TAKEAWAYS · CARRY THIS INTO EVERY CLASSIFICATION PROBLEM

# A decision framework for classification

**1**

Define the engineering meaning of each class

**2**

Inspect class balance before choosing metrics

**3**

Start from a baseline; stratify; tune only in training CV

**4**

Choose metrics from the consequences of each error

**5**

Select the model from the stated priority, not accuracy

**6**

Report limits and where the result can generalise

THE CENTRAL LESSON

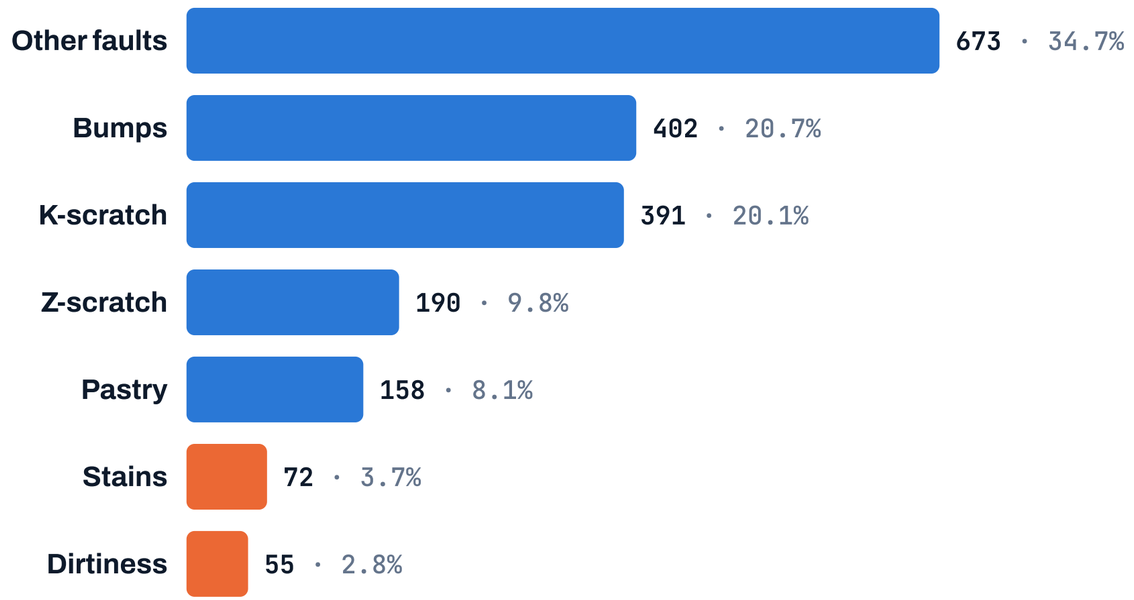
**High test accuracy is not proof of generalisation, or of engineering safety.**

EXTENSION · APPENDIX B NOTEBOOK · UCI STEEL PLATES FAULTS

# From binary to multiclass classification

## 1,941 steel plates, 7 mutually exclusive fault types

Orange: rare classes below 5%. The data we explored in Session 02.




**Binary**
stable vs unstable · 2 × 2 matrix

**Multiclass**
7 fault types · 7 × 7 matrix

NEW CONCEPTS

- Per-class precision, recall and F1
- **Macro** average: every class counts equally
- **Weighted** average: common classes count more
- Off-diagonal cells: which faults are confused

Same three model families, same stratified CV and test-set discipline. The evaluation becomes class-specific.

UCI Machine Learning Repository: Steel Plates Faults (Buscema, Terzi & Tastle, 2010), CC BY 4.0. doi:10.24432/C5J88N.

HOMEWORK · STEEL PLATE FAULT CLASSIFICATION

# Apply today’s workflow to seven classes

| | |
|---|---|
| 01 | Explore the data and plot the seven-class distribution |
| 02 | Make a stratified train/test split; keep the test set untouched |
| 03 | Fit a majority baseline, logistic regression, a decision tree and a random forest |
| 04 | Tune at least one model with stratified CV on the training data |
| 05 | Report the confusion matrix, balanced accuracy, per-class P/R/F1, macro-F1 and weighted-F1 |
| 06 | Explain which faults are confused, and choose a model with explicit criteria |



DELIVERABLES

- A runnable notebook
- A confusion-matrix plot
- A comparison table
- A short engineering interpretation

Start from the Appendix B notebook. A high test score alone does not make a model ready for production inspection.

OPTIONAL · FOR AFTER THE SESSION

# Two appendix notebooks

APPENDIX A

## How trees and forests learn

- Entropy, Gini impurity and information gain
- A decision tree’s boundary, step by step
- Bootstrap samples, feature randomness, voting
- Out-of-bag estimates and feature importance

`Appendix_A_Trees_Entropy_Information_Gain_Random_Forest.ipynb`

APPENDIX B

## Multiclass classification

- Steel Plates Faults: 1,941 plates, 7 classes
- A 7 × 7 confusion matrix
- Per-class, macro and weighted metrics
- The starting point for the homework

`Appendix_Multiclass_Classification_Steel_Plates_Faults.ipynb`



**Reference slides for Appendix A follow after the closing slides.**

# Questions

Unmute, or drop your question in the chat.



Logistic regression

Confusion matrix

Imbalance & thresholds

Slope stability

**Dr. Abdul Waris**
Research Associate · Universität Siegen, Germany

END OF SESSION 05

# Thank you.

See you in session 06: **Artificial Neural Networks.**

**Dr. Abdul Waris**
Research Associate · Universität Siegen, Germany

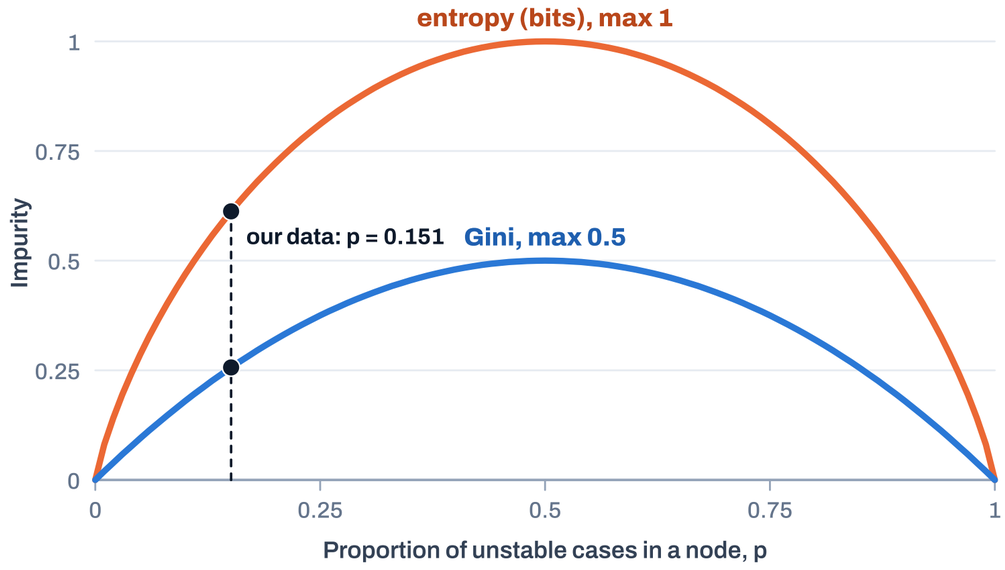
**Course page**
qudeltastudios.com/courses

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APPENDIX A1 · HOW MIXED IS A NODE?

# Entropy and Gini measure class impurity

## Both are 0 for a pure node, largest at 50 : 50




**ENTROPY (BITS)**

$$H = -\Sigma_k\, p_k \log_2 p_k$$

**GINI IMPURITY**

$$G = 1 - \Sigma_k\, p_k^{\,2} = 2p(1-p)$$

| Pure | Our data | 50 : 50 |
|---|---|---|
| H 0.000 | H 0.612 | H 1.000 |
| G 0.000 | G 0.256 | G 0.500 |

Different scales: compare impurity reductions with the same criterion, not raw values. Entropy describes a node's label mix, not physical uncertainty.

APPENDIX A2 · DOES A SPLIT REDUCE IMPURITY?

# Information gain compares candidate splits

PARENT IMPURITY MINUS SIZE-WEIGHTED CHILD IMPURITY

$$IG = H(\text{parent}) - \Sigma_j \frac{n_j}{n} H(\text{child}_j)$$

PARENT NODE: 100 CASES · 15 UNSTABLE / 85 STABLE · H = 0.610

## Split A

50 cases

10 / 40 · H 0.722

50 cases

5 / 45 · H 0.469

IG = 0.0144 bits · ΔGini 0.0050

## Split B · preferred

50 cases

12 / 38 · H 0.795

50 cases

3 / 47 · H 0.327

IG = 0.0486 bits · ΔGini 0.0162

Information gain scores one split at one node. Feature importance aggregates contributions across a whole model.

Illustrative counts from the Appendix A notebook, not measured from the 616-case dataset.

APPENDIX A3 · MATCH THE EXPLANATION TO THE CODE

# Entropy or Gini?

| CRITERION | BINARY FORMULA | INTUITION | SCIKIT-LEARN |
|---|---|---|---|
| **Entropy** | $-p \log_2 p - (1-p) \log_2 (1-p)$ | Information-theoretic impurity | `criterion="entropy"` |
| **Gini** | $2p(1 - p)$ | Chance of mislabelling a random case using the node's proportions | `criterion="gini"` · default |

**TEACHING RULE** Today's notebook uses the default **Gini** criterion. If you teach entropy, say so explicitly, or set `criterion="entropy"` and compare.

APPENDIX A4 · BAGGING PLUS RANDOM FEATURE SUBSETS

# Random forest: many decorrelated trees

**Training data**
`N rows`

| | |
|---|---|
| Bootstrap 1 | Tree · random features |
| Bootstrap 2 | Tree · random features |
| Bootstrap … | … |
| Bootstrap B | Tree · random features |

**Aggregate**
vote · average probability

## One bootstrap sample of 12 rows

`drawn: [1, 9, 7, 5, 5, 10, 1, 8, 2, 1, 6, 11]`




Different samples and feature randomness make trees less alike; averaging them reduces variance compared with one deep tree.

Bootstrap draw reproduced from the Appendix A notebook (random seed 42). Rows marked “out” are out-of-bag for this tree.

APPENDIX A5 · WHAT EACH SETTING CHANGES

# Random forest mechanics

### `n_estimators`

Number of trees. More trees stabilise predictions, at more compute.

### `max_features`

Features considered at each split. Controls how different the trees are.

### `max_depth · min_samples_leaf`

Limit tree complexity; can reduce overfitting.

### `class_weight`

Gives the rare class more weight. Check recall and precision, not accuracy.

### `oob_score`

Rows left out of a tree's bootstrap give an internal estimate. It does not replace a held-out test set.

Random forest is not automatically superior. Compare on the same held-out data, with metrics aligned to the application.

APPENDIX A6 · FURTHER READING AND PRACTICAL CAUTIONS

# Where to go next

### SCIKIT-LEARN DOCUMENTATION

Decision trees and impurity criteria
scikit-learn.org/stable/modules/tree.html

Random forests
scikit-learn.org/stable/modules/ensemble.html#forest

Classification metrics
scikit-learn.org/stable/modules/model_evaluation.html

### SLOPE-STABILITY DATASET CAVEAT

616 cases on a structured grid; labels from Morgenstern–Price FOS

A random split tests interpolation among grid points, not field-failure prediction

Hold out a parameter region for a stronger test of generalisation